Call for Papers

Eye Tracking and Brain–Computer Interfaces in AR/VR: Progresses, Challenges and Opportunities

Workshop at ACM UbiComp/ISWC 2026

October 11–12, 2026, Shanghai, China

Eye tracking and brain-computer interfaces are becoming increasingly important for ubiquitous, wearable, and immersive computing. Where people look can reveal visual attention, information needs, task strategies, and interaction intention. Brain signals can provide complementary information about cognitive load, affective state, attention, fatigue, and neural intention. Together, gaze and brain signals offer new opportunities for building adaptive, context-aware, and human-centered interactive systems.

This workshop aims to bring together researchers and practitioners from eye tracking, brain-computer interfaces, human-computer interaction, ubiquitous computing, wearable computing, extended reality, physiological sensing, and human-AI interaction. We invite contributions that explore how gaze and neural sensing can support future interactive systems in everyday, mobile, wearable, and immersive contexts.

The workshop will provide a forum for discussing current research, emerging systems, open challenges, and future directions. We particularly welcome contributions that connect eye tracking and/or BCI to UbiComp/ISWC themes, including wearable sensing, context-aware systems, real-world deployment, adaptive interaction, privacy, and social acceptability.

Topics of Interest

Topics include, but are not limited to:

- Gaze-based interaction for ubiquitous, wearable, and AR/VR systems
- Brain-computer interfaces for interaction and adaptive systems
- Multimodal fusion of gaze, EEG, physiological signals, body movement, and contextual data
- Attention, intention, cognitive load, fatigue, and affect recognition
- Gaze-BCI interaction techniques in AR, VR, and MR

- Wearable and mobile eye tracking
- Lightweight, mobile, and dry-electrode EEG systems
- Neuroadaptive and attention-aware interfaces
- Human-AI interaction using gaze and brain signals
- Collaborative and multi-user gaze/BCI systems
- Accessibility and assistive applications
- Applications in healthcare, education, training, work, and smart environments
- Datasets, benchmarks, toolkits, and evaluation methods
- Robustness, calibration, personalization, and long-term deployment
- Privacy, ethics, social acceptability, and responsible use of gaze and neural data
- Critical, speculative, and design-oriented perspectives on gaze and BCI in everyday computing

Submission Types

We invite the following types of submissions:

- **Position papers**: 2–4 pages presenting viewpoints, challenges, or future research directions.
- **Work-in-progress papers**: 2–4 pages describing ongoing research, early findings, systems, or studies.
- **Demo proposals**: 2–4 pages describing interactive prototypes, toolkits, datasets, or sensing systems.
- **Provocation papers**: 1–2 pages presenting critical, speculative, or discussion-provoking ideas.

Submissions may focus on eye tracking, BCI, or the combination of both. We also welcome submissions that focus on broader multimodal sensing if they clearly connect to gaze, neural sensing, or adaptive interaction.

Submission Format

Submissions should follow the official UbiComp/ISWC 2026 workshop paper template and formatting instructions.

Accepted workshop papers are planned to be included in the ACM Digital Library and the UbiComp/ISWC 2026 adjunct proceedings, subject to the official publication requirements of the conference.

Submissions should be made through PCS:
Submission link: TBD

Review Process

Each submission will be reviewed by at least two reviewers. Submissions will be evaluated based on:

- relevance to the workshop theme;
- originality and significance;
- clarity of presentation;
- quality of contribution;
- potential to stimulate discussion;
- diversity of perspectives and methods.

Authors of accepted submissions will be invited to present their work at the workshop. At least one author of each accepted submission must register for and attend the workshop in person.

Important Dates

All deadlines are 23:59 AoE.

- **Call for Papers released**: May 5, 2026
- **Submission deadline**: TBD
- **Notification to authors**: TBD
- **Camera-ready deadline**: TBD
- **Workshop**: October 11 or 12, 2026
- **Main conference**: October 13–15, 2026

Workshop Format

The workshop will be a half-day in-person event. It will include short presentations, invited perspectives, breakout discussions, and group synthesis. The goal is not only to present accepted papers, but also to build a shared research agenda for gaze and BCI in ubiquitous and wearable interaction.

Authors of accepted submissions will be invited to present their work. In addition, the workshop will be open to registered UbiComp/ISWC attendees who are interested in the topic, subject to room capacity. This format will ensure high-quality focused discussions while also allowing broader participation from the UbiComp/ISWC community.

Organizers

- Shiwei Cheng, Zhejiang University of Technology
- Jinghui Hu, Lancaster University
- DONG Zhanxun, Shanghai Jiao Tong University

Contact

For questions, please contact:
TBD workshop email

Workshop website:
TBD